ECE 220 Computer Systems & Programming

Lecture 17 — Linked Lists

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Prof. Sayan Mitra mitras@illinois.edu
Course website: courses.grainger.illinois.edu/ece220/fa2026

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Electrical & Computer Engineering
GRAINGER COLLEGE OF ENGINEERING

A Late Enrollment

```
int insert_sorted(student ece220[], int n,
                  student s) {
    int i = n;
    while (i > 0 && ece220[i - 1].UIN > s.UIN) {
        ece220[i] = ece220[i - 1];  /* move up */
        i--;
    }
    ece220[i] = s;
    return n - i;
}
int main() {
    student ece220[200] = {
        {"Jamie", 123456789, 3.89},
        {"Robin", 246813579, 3.52},
        {"Alex",  987654321, 3.98}};
    student casey = {"Casey", 135792468, 3.71};
    int n = 3, moved, i;

    moved = insert_sorted(ece220, n, casey);
    n++;
```

```
Jamie 123456789 3.89
_____________________
Robin 246813579 3.52
Alex 987654321 3.98
___ records moved, 108 bytes each
```

200 students, and a UIN below all of them: _____ records, _____ bytes copied, for one new student.

A linked list adds Casey by changing two pointers. No record moves.

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        ece220[i] = ece220[i - 1];  /* move up */
        i--;
    }
    ece220[i] = s;
    return n - i;
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int main() {
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    int n = 3, moved, i;

    moved = insert_sorted(ece220, n, casey);
    n++;
```

ece220 is sorted by UIN. Casey, UIN 135792468, enrolls after the roster was built. The program prints

```
Jamie 123456789 3.89
_____________________
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___ records moved, 108 bytes each
```

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Linked List Data Structure

1. A **data structure** defines how data is organized and accessed.

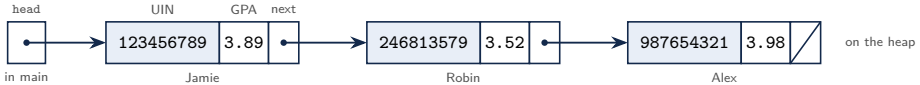
2. A data structure comes hand in hand with the **algorithms** that create it, access it, operate on it, and destroy it.

3. **What is a linked list, and what are the algorithms that create one, access it, operate on it, and destroy it?**

`find_node`, `add_node` and `remove_node` are each recursive: a base case, then the same call on the rest of the list.

The Linked List Data Structure

A **linked list** is an ordered collection of **nodes**, each holding some data, joined by pointers: each node holds the address of the next node.




```
typedef struct studentStruct Node;
struct studentStruct {
    int   UIN;
    float GPA;
    Node *next;
};
```

The `typedef` comes first because the struct uses the name `Node` inside its own definition.

- The first node is the _____; the variable `head` holds _____
- The last node is the _____; its `next` is _____, drawn as a slash
- The empty list: `head ==` _____
- `sizeof(Node)` on the laptop: _____
- The nodes live _____; `head` lives in `main`'s activation record (AR) on the run-time stack (RTS)

Array vs. Linked List

	Array `student ece220[200]`	**Linked list** `Node *head`
Memory allocation		
Memory structure		
Memory overhead		
Order of access		
Insertion / deletion		

Binary search needs random access. A sorted linked list is searched from the head, one node at a time.

Build a List by Hand

```
Node *head, *n1, *n2, *n3;

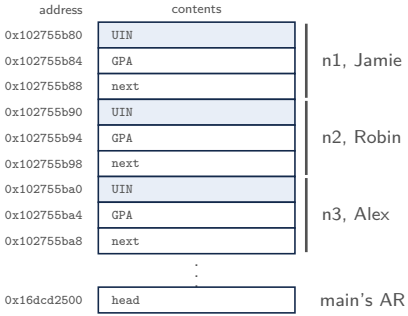
n1 = (Node *)malloc(sizeof(Node)); /* Jamie */
n2 = (Node *)malloc(sizeof(Node)); /* Robin */
n3 = (Node *)malloc(sizeof(Node)); /* Alex  */
if (n1 == NULL || n2 == NULL || n3 == NULL)
    return 1;
n1->UIN = 123456789;  n1->GPA = 3.89;
n2->UIN = 246813579;  n2->GPA = 3.52;
n3->UIN = 987654321;  n3->GPA = 3.98;
n1->next = n2;
n2->next = n3;
n3->next = NULL;
head = n1;

print_list(head);
printf("second GPA: %.2f\n", head->next->GPA);
```

- `head->UIN` is _____
- `head->next->next->GPA` is _____
- `head->next->next->next` is _____; the last line prints _____

Draw `head` and the three nodes after `head = n1;`

The List in Memory



Lower addresses at the top. One run of `build.c` on a Mac; the addresses change from run to run.

`build.c` prints the addresses:

```
sizeof(Node) = 16
n1 at 0x102755b80  next _____________
n2 at 0x102755b90  next _____________
n3 at 0x102755ba0  next _____________
head at 0x16dcd2500 holds _____________
```

- One node is 16 bytes: `UIN` at +0, `GPA` at +4, `next` at _____
- `n1->next` holds _____: a `next` field is an address, as `head` is
- `n2->GPA` is at _____
- `malloc` placed these nodes 16 bytes apart. After a `free` and a new `malloc`, a node can land anywhere; `next` alone keeps the order.

Walk the List: `print_list`, Two Ways

With a loop (`build.c`)

```
void print_list(Node *head) {
    Node *current = head;
    while (current != NULL) {
        printf("%d %.2f\n",
               current->UIN, current->GPA);
        current = current->next;
    }
}
```

- `current` starts at _____; the loop runs while _____
- One step: _____
- `head` itself _____: `main` still has the list

Walk the List: `print_list`, Two Ways

With a loop (`build.c`)

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void print_list(Node *head) {
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    }
}
```

- `current` starts at _____; the loop runs while _____
- One step: _____
- `head` itself _____: `main` still has the list

Recursively (`llist.c`)

```
void print_list(Node *head) {
    if (head == NULL)
        return;
    printf("%d %.2f\n",
           head->UIN, head->GPA);
    print_list(head->next);
}
```

- Base case: _____, the empty list
- Recursive case: print this node, then the same call on _____
- Calls for the three-node list: _____

A list is one node followed by a shorter list, or NULL. Every function today has that shape.

Both print `123456789 3.89`, `246813579 3.52`, `987654321 3.98`, one per line.

find_node: Search the Sorted List

```
Node *find_node(Node *head, int S_UIN) {
    /* base case 1: list end, or past S_UIN */
    if (head == NULL || head->UIN > S_UIN) {
        printf("record not found\n");
        return NULL;
    }
    /* base case 2: this node matches S_UIN */
    if (head->UIN == S_UIN) {
        printf("record found\n");
        return head;
    }
    /* recursive case: the remaining list */
    return find_node(head->next, S_UIN);
}
```

It returns the node's address, so `main` can read or change the record: `p = find_node(head, 246813579);` then `p->GPA`.

Trace `find_node(head, 246813579)`

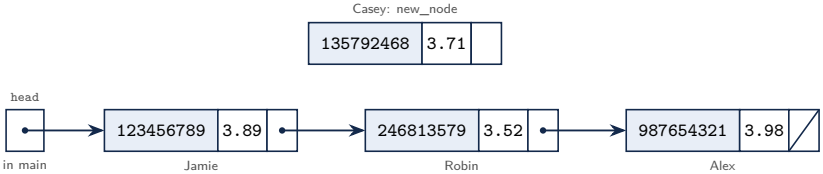
- call 1: `head` is Jamie's node; 123456789 $<$ 246813579: _____
- call 2: `head` is Robin's node: _____; it returns _____
- call 1 returns _____

Trace `find_node(head, 200000000)`

- call 2: 246813579 $>$ 200000000: _____; Alex's node is _____
- A UIN above every UIN in a list of n nodes: _____ calls

Add Casey: Which Pointers Change?

`add_node(&head, 135792468, 3.71)`: Casey's UIN falls between Jamie's and Robin's. Draw the pointers after the call.



- 1: `new_node->next =` _____
- 2: _____
- Step 1 comes first: otherwise _____

- A UIN below 123456789 goes before Jamie. The pointer that changes is _____
- So `main` passes `&head`. `Node **list` holds the address of a pointer: `main`'s `head` on the first call, _____ on each recursive call. `*list = new_node` writes into whichever it is.

In-Class Problem: `add_node`

Base case 1 covers the empty list, a UIN below the first node's, and every spot in the middle: the new node goes in front of `*list`.

- Its test: `*list == NULL ||` _____
- The UIN exists: _____
- The recursive call passes _____
- `main` adds Alex, Jamie, Robin in that order; then `head` points at _____

```
void add_node(Node **list,
              int new_UIN, float new_GPA) {
    Node *new_node;
    /* base case 1: insert new node at 'head' */



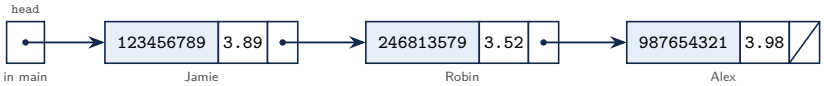

    /* base case 2: UIN already exists */


    /* recursive case: the remaining list */


}
```

Remove Robin: Which Pointer Changes?

`remove_node(&head, 246813579)`. Draw the pointers after the call, and mark the node that is freed.



- The pointer that changes: _____; its new value: _____
- Then _____. Its address is saved in `old` first, because afterwards _____

- Remove Jamie instead: the pointer that changes is _____. The same two lines do it, because `*list` is `head` on the first call
- `remove_node(&head, 200000000)`: the walk stops at _____; nothing is freed

In-Class Problem: `remove_node`

- Base case 1 is `find_node`'s test, written with `*list`: `*list == NULL || _____`
- Base case 2, in order: save _____ in `old`; set `*list` to _____; then _____
- `main` removes Robin twice. The second call prints _____
- `free_list` frees _____, then this node: after `free(head)`, `head->next` must not be read

```
void remove_node(Node **list, int old_UIN) {
    Node *old;
    /* base case 1: list end, or past old_UIN */




    /* base case 2: found: unlink and free it */




    /* recursive case: the remaining list */


}
```

Summary & What's Next

Today

- **The problem**: a late enrollment in a sorted array shifts every record after it
- **Nodes and links**: `Node` holds `UIN`, `GPA` and `next`; `head` holds the first node's address; the tail's `next` is `NULL`; one `malloc` per node, nodes anywhere on the heap, 8 of 16 bytes for the pointer
- **Build and walk**: three `malloc`s and three `next` assignments; `print_list` with `current = current->next`, or recursively on `head->next`
- **find_node**: base case 1 (list end, or past `S_UIN`: NULL), base case 2 (match: this node), else the same call on `head->next`
- **add_node** and **remove_node**: `Node **list` holds the address of `head` or of a node's `next`; inserting sets two pointers, removing sets one and frees; no record moves; `leaks` reports 0

Next lecture

- Problem solving with linked lists: `add_node(&head, ...)` drawn on the RTS; doubly linked lists; a stack and a queue on a linked list

Code from today: `insert_array.c`, `build.c`, `llist.c` (skeleton `llist_skel.c`).